

Logic Circuits · AS · Practice

U03b · 24 written + 6 pen-first = 30 marks · Name: _____

Questions labelled Ours are authored. Preserve supplied expressions and circuit structure. NOT has one input; all other gates have two.

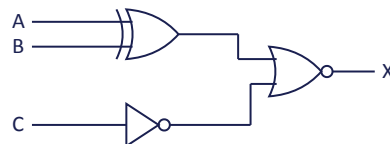
1. Ours: from the requirement

An aircraft warning horn H sounds if $G=0$ (landing gear not locked) while $L=1$ (low altitude), or if $T=1$ (test button pressed). Write the expression for H. [2]

2. Ours: expression into wires

Draw the circuit for $X = (A \text{ XOR } B) \text{ NAND } (B \text{ OR } C)$. Label inputs and output; make connections clear. [3]

3. Ours: read the circuit



Write the logic expression for X, following the circuit shown. [2]

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4. Ours: follow every row

Complete the output column for $X = (A \text{ NAND } B) \text{ AND } (A \text{ OR } C)$. Working columns are optional.

[4]

A	B	C	A NAND B	A OR C	X
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

5. Ours: outputs back to an expression

For inputs AB in row order **00, 01, 10, 11**, the outputs are **1, 0, 0, 1**.

Write a correct expression using the six syllabus gates. No simplification is required.

[2]

6. Ours: read the wiring

State what a bubble at a gate output means. State what a wire crossing with no junction dot means in these diagrams.

[2]

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7. Ours: exam-shape rehearsal

New circuits, modelled on Cambridge drawing and table question shapes. These are authored questions, not past-paper extracts.

(a) Draw the circuit for $Z = (P \text{ OR } Q) \text{ NOR } (R \text{ NAND } S)$. Preserve this structure; all gates have at most two inputs. [2]

(b) Complete the output column for $Z = (\text{NOT } P \text{ AND } Q) \text{ OR } (R \text{ XOR } Q)$. [2]

P	Q	R	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

8. Ours: test a proposed replacement

(a) Complete $U = A \text{ XOR } B$ and $V = A \text{ OR } B$. [4]

A	B	U	V
0	0		
0	1		
1	0		
1	1		

(b) Give the input row that disproves $U = V$ for all inputs. [1]

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9. Ours: match the fingerprint · pen first [6]

 **Paper first, then Circuit Forge.** Two slots are wired as $X = (A ? B) ? C$. Choose AND, OR, XOR, NAND or NOR for each slot. The target outputs in row order **000 to 111** are **0, 0, 0, 1, 0, 1, 0, 1**. All six marks are human marked.

(a) Choose the two gates and sketch the circuit with its labels.

[2]



(b) For your circuit, calculate X for inputs 001 and 110. Show the intermediate gate output in each case.

[2]

(c) A classmate proposes $(A \text{ AND } B) \text{ OR } C$. Give one input combination where it differs from the target, and give both outputs.

[2]

Keyboard companion: Karel 11 Spiral and 12 StoneMason remain the early-course programming tasks, outside these 30 marks. No Python file is needed for this gate investigation.

Above and beyond · unmarked, for both groups. A half-adder uses XOR for Sum and AND for Carry. Explain why two outputs are needed when adding $1+1$. Why does a Boolean truth table alone not tell you how quickly a real adder responds?

Additional space - label the question number.
